

5-7-1: Examining VPNs and QoS

At the end of this episode, I will be able to:

1. Identify and explain the purpose of Quality of Service (QoS), given a scenario.

Learner Objective: *Identify and explain the purpose of Quality of Service (QoS), given a scenario*

Description: In this episode, the learner will examine Virtual Private Networks (VPNs) and concepts such as VPN types, protocols, and security considerations. We will explore Quality of Service (QoS), 802.1Q, priority tagging and more..

- What are VPNs?
 - o A secure, encrypted connection over a public networks, such as the Internet
 - o Provide secure remote access or connect multiple sites.
- What are the different types of VPNs?
 - o Site-to-Site
 - ♣ Used to connect entire networks to each other, often employed for connecting remote offices to a company's main network.
 - ♣ Facilitates secure communication between different sites over the internet.
 - o Remote Access VPNs
 - ♣ Allow individual users to connect to a private network from a remote location.
 - ♣ Commonly utilized for telecommuting or accessing corporate resources remotely.
- What are VPN tunneling and security technologies bein used?
 - o PPTP \ (Point-to-Point Tunneling Protocol)
 - ♣ Older VPN protocol
 - ♣ Easy setup
 - ♣ Has security vulnerabilities
 - ♣ uses TCP port 1723 tunnel maintenance and IP protocol 47 for GRE (Generic Routing Encapsulation).
 - o L2TP (Layer 2 Tunneling Protocol)
 - ♣ PPTP (Microsoft) + L2F (Cisco) = L2TP
 - ♣ Commonly used with IPsec for encryption
 - ♣ Uses UDP ports 1701 for L2TP traffic, 500 and 4500 for IPsec
 - o IKEv2 (Internet Key Exchange version 2)
 - ♣ Part of the IPsec suite
 - ♣ Used for establishing and managing security associations.
 - ♣ Uses UDP ports 500 for the initial key exchange and 4500 for NAT traversal.
 - o MPPE (Microsoft Point-to-Point Encryption) - Works with PPTP for encrypting data on Microsoft VPN networks.
 - o SSTP (Secure Socket Tunneling Protocol)
 - ♣ Developed by Microsoft, uses SSL/TLS for encryption
 - ♣ Offers security and ability to bypass most firewalls.
 - ♣ Uses TCP port 443, the same port used for HTTPS traffic.
 - o IPSec (Internet Protocol Security) - A suite of protocols for securing internet protocol communications by authenticating and encrypting each IP packet.
- What is QoS?
 - o A set of technologies used to:
 - ♣ Manage network traffic
 - ♣ Prioritize different types of data
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- What is the purpose of QoS
 - o Ensures predictable network behavior and performance
 - o Particularly for time-sensitive applications like VoIP and video conferencing.
 - o Aims for reliable and consistent data packet delivery
 - o The foreduced latency, jitter, and packet loss.
- Where do we see this implemented or used?
 - o In network devices such as routers and switches.
- What do these QoS functions perform?
 - o Implements traffic policies - shaping and policing
 - o Bandwidth allocation - determining available bandwidth for traffic.
 - o Queuing - prevents traffic congestion
- How can routers determine priority?
 - o Terms of Service (ToS) Field in IPv4
 - ♣ An 8-bit field in the IPv4 header for indicating packet priority.
 - ♣ Includes bits for precedence, delay, throughput, and reliability.
 - o Differential Service Code Point (DSCP)
 - ♣ Occupies the first 6 bits of the ToS field in IPv4/Traffic Class field in IPv6.
 - ♣ Allows for 64 different classes of service
 - ♣ Provides granular packet classification.
- How can the switches determine priority?
 - o VLAN tagging (IEEE 802.1q)
 - ♣ Includes priority bits in the VLAN tag
 - ♣ Allow QoS policies based on VLAN ID
 - ♣ Priority level (802.1p)
 - o IEEE 802.1p (Priority Tagging)
 - ♣ Adds a 3-bit class of service (CoS) priority field in the Ethernet frame
 - ♣ Defines 8 priority levels
 - ♣ Used by switches for traffic prioritization (forwarding decisions)
 - ♣ Used by switches for congestion management.

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- Additional Resources:
 - o If applicable